

Citation Integrity Audit

Prompt Sensitivity and Its Effect on the Stability of LLM-Generated Research Conclusions

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Programme: IT-HMIIT

AI Tool: Claude

Model/Version: Opus-5

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Purpose

This standalone report records the citation-authenticity and claim-citation-support audit performed on a systematic sample of the AI-generated research paper. The audit results are based on the completed Lab 1 work.

Paper profile

Measure	Recorded result
Paper title	Prompt Sensitivity and Its Effect on the Stability of LLM-Generated Research Conclusions
Approximate pages	13
Approximate word count	6,307 (5,023 excluding references)
Number of sections	11
Total references	28
Approximate in-text citation occurrences	99
Identifiers supplied	28 DOI; 0 PMID-only; 0 arXiv-only; 0 URL-only; 0 without persistent identifier

Systematic sample

For 28 references, the Lab 1 procedure selected the first 3, last 3, and 4 approximately evenly distributed middle references.

Audit slot	Original bibliography reference no.
First 3 references	R1, R2, R3
Middle references	R4, R5, R6, R7
Last 3 references	R8, R9, R10

Pre-verification plausibility

Ref.	Recorded initial impression	Plausibility
R1	Landmark paper; author list and venue expected	5
R2	Well-known survey; detailed ACM metadata	5
R3	Recent arXiv-only survey; plausible but unfamiliar	4
R4	Familiar ICML paper; clean identifier	5
R5	Findings of ACL; exact page range	4
R6	Distinctive title; known authors; NeurIPS venue	4
R7	ArXiv preprint; ten authors; provocative title	3
R8	Nature article; volume/page information	5
R9	Nature Reviews Physics comment piece	4
R10	Nature news feature with survey size	4

Authenticity and metadata verification

The audit checked existence, title, authors, year, venue/source, supplied bibliographic details and whether the supplied identifier belonged to the cited publication.

Ref.	Verification source(s)	Identifier	What was found
R1	arXiv; NeurIPS proceedings; Semantic Scholar	arXiv:2005.14165	Exists; 31 authors; NeurIPS vol. 33, pp. 1877–1901, 2020
R2	ACM Digital Library via DOI; DBLP; Semantic Scholar API	DOI 10.1145/3560815	Exists; ACM CSUR 55(9), Article 195, 2023; six authors
R3	arXiv abstract page	arXiv:2402.07927	Exists; title, six authors and 2024 date match; cited as preprint
R4	arXiv; ICML 2021 proceedings	arXiv:2102.09690	Exists; five authors; ICML 2021; identifier resolves
R5	ACL Anthology	DOI 10.18653/v1/2024.findings-acl.375	Exists; three authors; Findings of ACL 2024, pp. 6287–6310
R6	arXiv; NeurIPS 2023 record	arXiv:2305.04388	Exists; four authors; NeurIPS 2023; resolves correctly
R7	arXiv abstract page	arXiv:2305.17926	Exists; all ten authors match; later peer-reviewed ACL version noted
R8	Nature via DOI	DOI 10.1038/s41586-024-07146-0	Exists; Nature 627, pp. 49–58, 2024; both authors match
R9	Nature Reviews Physics	DOI 10.1038/s42254-023-00581-4	Exists; vol. 5, pp. 277–280, 2023; four authors match
R10	Nature	DOI 10.1038/d41586-023-02980-0	Exists; Nature 621, pp. 672–675, 2023; news feature

Final classifications

Ref.	Code	One-line reason
R1	A	Title, authors, year, venue and pages match the NeurIPS proceedings record.
R2	A	ACM volume, issue, article number and page range confirmed via ACM/DBLP.
R3	A	Title, six authors and 2024 date match the arXiv record.
R4	A	arXiv identifier resolves to the cited ICML 2021 paper and author list.
R5	A	ACL Anthology confirms authors, venue and pages 6287–6310.
R6	A	arXiv identifier resolves to the cited NeurIPS 2023 paper.
R7	A	All ten authors and identifier match; later ACL version also noted.
R8	A	DOI resolves to Nature 627:49–58 with the cited authors.
R9	A	DOI resolves to Nature Reviews Physics 5:277–280 with the cited authors.
R10	A	DOI resolves to Nature 621:672–675 with the cited authors; correctly treated as a news feature.

Authenticity score

A = 10, B = 0, C = 0, D = 0, E = 0. Authenticity points = 4(10) + 3(0) + 1(0) + 1(0) = 40; maximum = 40. Authenticity Score = 100/100 (Excellent).

Prediction accuracy

Correct genuine/fake predictions = 10 out of 10. Prediction Accuracy = 100%.

Claim-citation support audit

For each genuine reference, the cited work was compared with the sentence to which the AI attached the citation. Codes: S = fully supported; P = partially supported, over-generalised or imprecisely worded; N = not supported; U = unable to determine.

Ref.	Code	Evidence / reasoning
R1	S	Central claim of the paper; abstract states tasks are specified through text interaction without gradient updates.
R2	S	Survey states the pre-train-prompt-predict paradigm shift.
R3	S	Abstract describes systematic categorisation by application area with datasets and models.
R4	S	Abstract reports prompt-format/example/order instability and gains from contextual calibration.
R5	S	Source's stated motivation is template-controlled evaluation because effects may not survive template changes.
R6	S	Source reports plausible reasoning traces can hide a biasing feature, with accuracy drops up to 36%.
R7	S	Source reports rankings can be altered by changing candidate presentation order.
R8	P	Source discusses illusions of understanding/explanatory depth; the AI wording used "explanatory breadth," conflating distinct constructs.
R9	S	Source calls for careful, constrained and transparent use in scientific practice.
R10	S	Nature feature reports a survey of more than 1,600 researchers and mixed views on AI in research.

Claim-support score

S = 9, P = 1, N = 0, U = 0 across 10 audited references. Support points = $2(9) + 1(1) = 19$. Claim-Support Score = 95/100.

Key finding

The single partial-support case was the Messeri and Crockett citation. The source supported a nearby idea involving illusion of understanding/explanatory depth, but the AI wording used "explanatory breadth." The audit therefore judged the claim partially supported because a term of art had drifted. Metadata and the DOI were correct.

Interpretation and limitation

The 100/100 authenticity score applies only to the 10-reference sample from the 28-reference paper. It does not establish that all 28 references are genuine. The Lab 1 record also notes that all 28 references carried DOI identifiers, including 13 arXiv preprint DOIs, and that the presence of a DOI does not by itself establish peer review.

Final lesson

Existence and claim support are different questions. Bibliographic verification can establish whether a publication exists and whether its metadata match; claim-support verification requires reading the source and comparing it with the exact research statement.